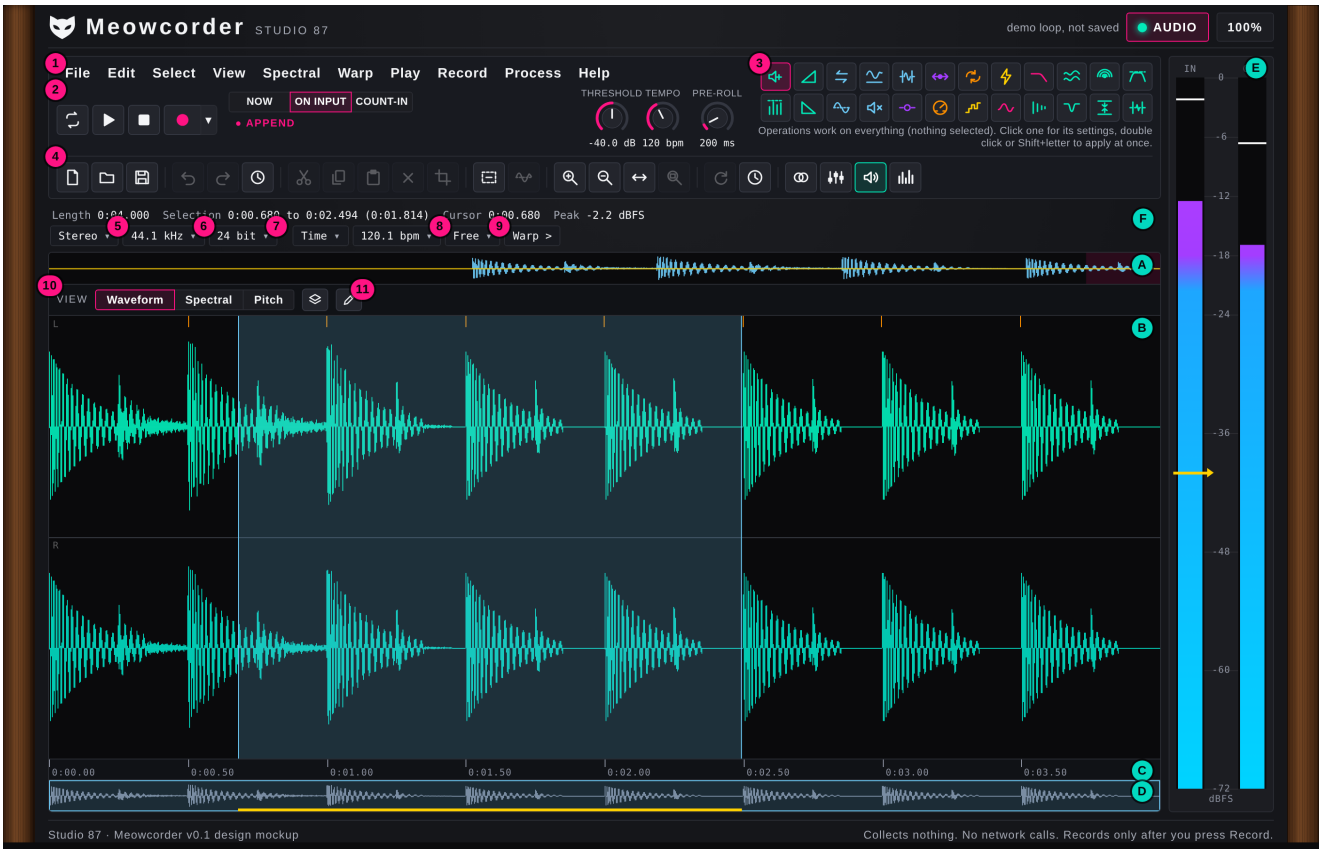
Meowcorder

Record, cut and reshape sound. Part of the Studio 87 cat plugin suite.



Free and open source, AGPL-3.0-or-later. Collects nothing, makes no network calls.

Intro

Meowcorder is a free recorder and sound editor for your music projects, made for Linux first and also for Windows and macOS. It is always listening to its input, so you can grab a sound you just heard, record a new one, cut and shape it, and drop it back into your project. It is part of the Studio 87 cat plugin suite, free and open source (AGPL-3.0-or-later).

Meowcorder has its own history of steps, separate from your DAW. You can undo, redo and repeat inside Meowcorder as much as you like without changing anything in your project until you drop the result back.

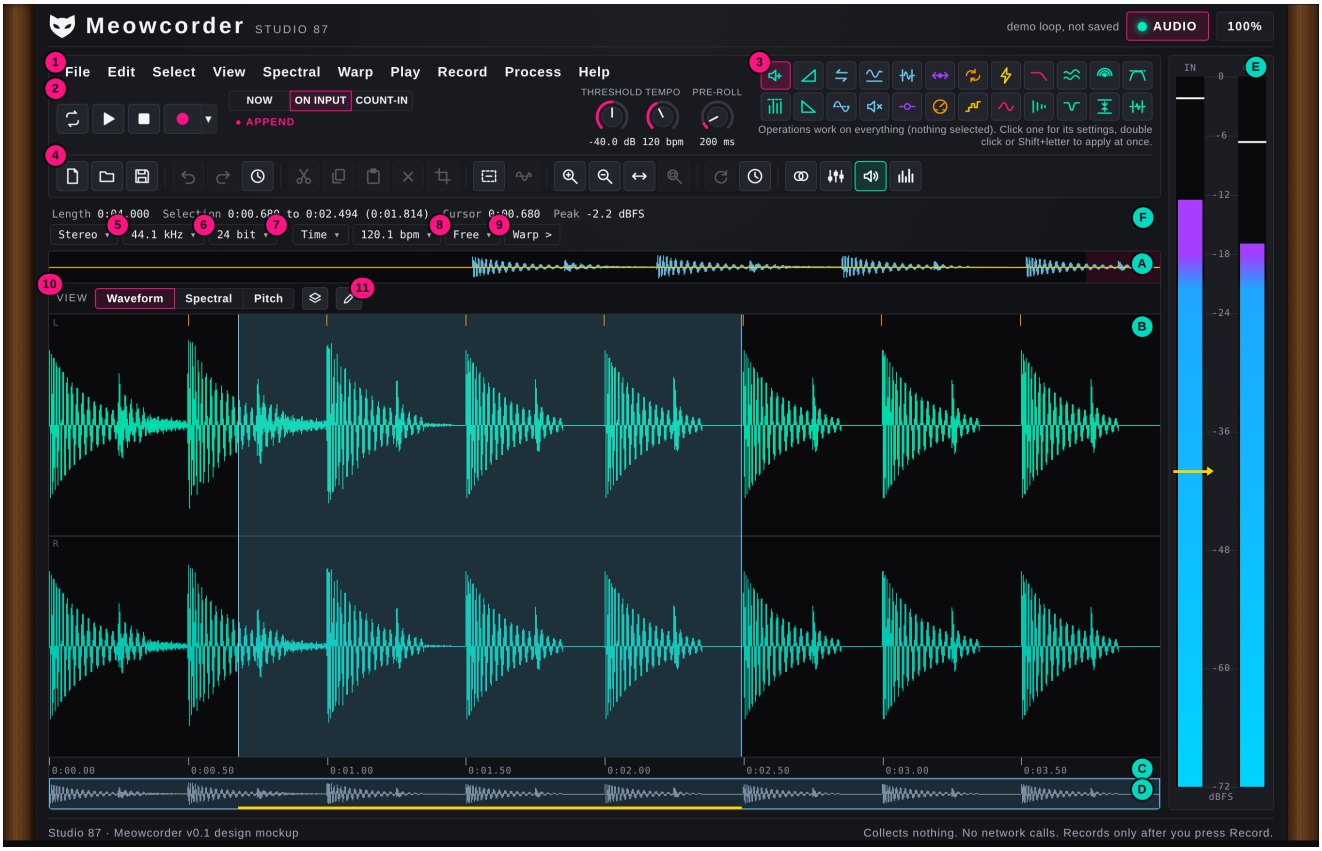
This manual describes the interface as designed in the interactive mockup. The plugin window is built from that mockup, with the same layout, controls and info bubbles.

Quick start

- Press Audio (top right) to start listening. The strip above the display shows the last 3 seconds of input, and the meters on the right show the level.
- Choose how a recording starts with the Now, On input and Count-in buttons, then press the pink record button. Press it again to stop.
- Drag over the waveform to select a part. Operations, cut, copy and paste work on the selection. With nothing selected they work on the whole sound.
- Click an operation (the colour grid at the top right) to open its settings, then press Apply. Double click an operation to apply it at once with its last settings.
- Press R to repeat the last operation, Ctrl+Z to undo, and Ctrl+Shift+Z to redo, or to apply the last action again when there is nothing to redo.
- Hover any control for one second to see what it does.

Interface tour

Pink numbers mark controls, teal letters mark displays. Every control has an info bubble: hover it for one second.



	What it is
1	Menu bar: every action with its shortcut
2	Transport and recording: loop, play, stop, record, start mode, take mode, Threshold, Tempo, Pre-roll
3	Operations: 24 effects and edits
4	Tool bar: file, undo, redo, history, edit, zoom, repeat, grab, input
5	Channels: what the sound is made of, click to change
6	Sample rate: click to convert
7	Bit depth: click to change
8	Sample tempo: click to set
9	Sync: free or synced to the host, warp mode
10	View bar: Waveform, Spectral, Pitch, layers
11	Waveform pencil
A	Live input: the last 3 seconds, pink band is the pre-roll
B	Main display: the view you are working in
C	Time line
D	Whole sound: drag the light box to scroll
E	Level meters: input and output, with the trigger arrow
F	Readout: length, selection, cursor, peak

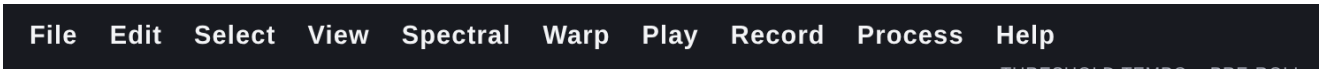
Signal flow: input (mixer channel or microphone) > always-listening ring buffer > recording > editor (selection, operations, views, warp) > playback or drag back into the project.

Header



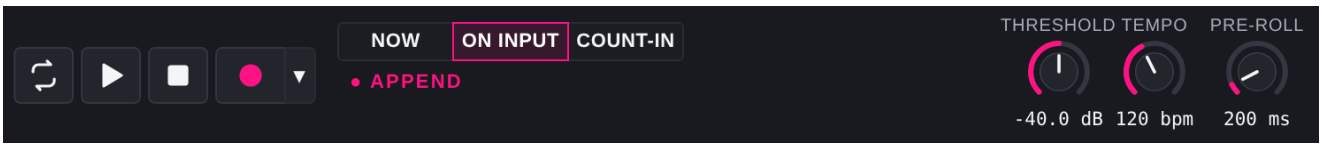
Control	Shortcut	What it does
Audio		Audio on or off. Browsers need one click before they play sound.
100%		Window size, percent of the default. Drag any edge or corner to resize; double click the wood to go back to 100%.

Menu bar



Control	Shortcut	What it does
File		New empty sound, Open file..., Save as WAV, Save bit depth, Audio on or off, Demo sounds
Edit		Undo, Redo, Redo, or repeat the last operation, Cut, Copy, Paste, Delete selection, Crop to selection, Repeat last operation, History...
Select		All, None, From cursor to the end, From the start to the cursor, Snap edges to zero crossings
View		Waveform view, Spectral view, Pitch view (piano roll), Waveform pencil (draw samples), Show views together, Together as, Switch overlay and split, Bring the next view to the front, Zoom in, Zoom out, Show the whole sound, Zoom to the selection, Time line unit, Next time line unit, Hit marks, Info bubbles, Power saving, Window size
Spectral		Show the spectral view, Tool, Amp amount, Eraser brush, Cut the box, Amp the box, Copy the box, Paste at the cursor, Clear the box
Warp		Warp, tempo and pitch..., Sync to the host tempo, Warp mode, Formants, Transpose, Bake the warp into the sound, Snap the notes to semitones, Set the tempo of this sound..., Detect the tempo again
Play		Play, Space bar rule, Play, Shift+Space rule, Play, Play button rule, Loop, Preview the chosen operation, Cursor to the start, Cursor to the end, Play keys settings...
Record		Record or stop, Grab the last seconds, Monitor the input, Record in mono, Start, Count-in length, Tempo, Click, A new take will, Threshold level, Pre-roll (on input), Input, Use the microphone instead of the mixer channel
Process		Apply the chosen operation, Level, Edit, Stereo, Time, Colour, Tone, Space, Dynamics
Help		Guide, Keyboard shortcuts, Settings, About & Support

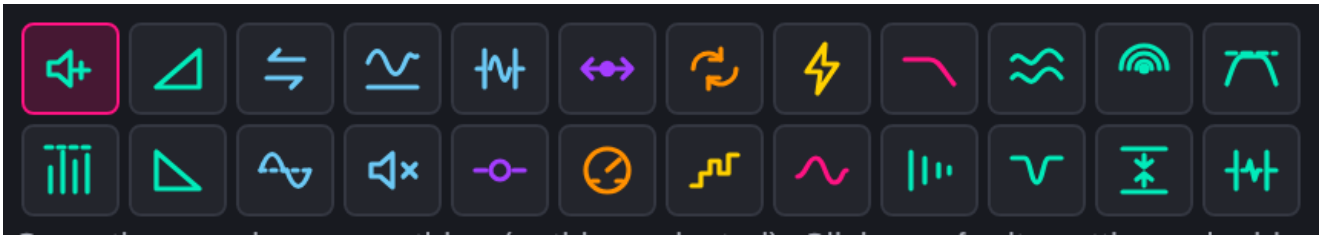
Transport and recording



Control	Shortcut	What it does
Loop	L	Repeats the playback again and again.
Play		Plays from the cursor, or the start of the selection. Pressed again while playing: stops. Change this in Settings.
Stop		Stops playing and recording.
Record	Enter	Records what comes in from the input. Where it starts is set by the buttons next to it.
Now		Recording starts the moment you press Record. Key 1.

Control	Shortcut	What it does
On input		Press Record to arm. Recording starts when the incoming sound passes the threshold; the pre-roll is included. Key 2.
Count-in		Press Record, hear the click for the chosen number of bars (4/4), and recording starts on the next downbeat. Key 3.
Append		What a new recording does: replace the sound, append at the end, punch over the selection or insert at the cursor. Click to change.
Threshold	-80.0 dB to 0.0 dB, default -40.0 dB	Level that starts a recording when Start is on Threshold.
Tempo	40 bpm to 240 bpm, default 120 bpm	Tempo of the click for the count-in and for recording with the click on. 4/4.
Pre-roll	0.0 ms to 3.00 s, default 200 ms	How much sound from before the start is included in the recording.

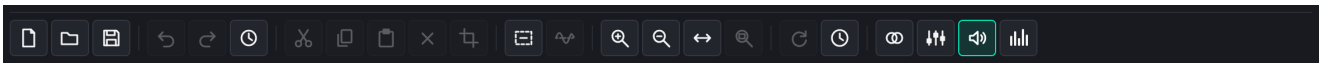
Operations



Control	Shortcut	What it does
Gain	Shift+G	Makes the selection louder or quieter by a fixed amount.
Normalize	Shift+N	Sets the level of the selection to a target. Never pushes the peak above 0 dBFS.
Fade in	Shift+I	Starts silent and rises to full level across the selection.
Fade out	Shift+O	Falls from full level to silence across the selection.
Reverse	Shift+R	Plays the selection backwards.
Invert polarity	Shift+P	Flips the wave upside down. Sounds the same alone, cancels against the original.
Remove DC offset	Shift+D	Centres the wave on zero. Removes a constant push that wastes headroom.
Silence	Shift+S	Replaces the selection with silence, same length.
Trim silence	Shift+X	Cuts quiet parts off the start and end.
Make mono	Shift+M	Turns the selection into one channel on both sides.
Balance	Shift+B	Moves the stereo image left or right. The quieter side goes down, the other stays.
Pitch and speed	Shift+Y	Plays the selection faster or slower. Pitch and length change together, like tape.
Make loop	Shift+J	Blends the end into the start so the selection repeats without a click. Becomes a little shorter.
Bitcrusher	Shift+K	Lo-fi: fewer bits and a lower sample rate.
Drive	Shift+V	Three analog style stages, the same as PurrOne: Drive, Transistor and Fuzz. Level matched.
EQ	Shift+E	Low shelf at 120 Hz, a bell and a high shelf at 8 kHz.
Filter	Shift+F	Low pass, high pass or band pass.
Delay	Shift+L	Echoes. The tail can be added after the selection so it rings out.
Chorus	Shift+H	Thickens by mixing in slightly detuned copies.
Phaser	Shift+A	Sweeping notches that move up and down.
Reverb	Shift+W	A room around the sound. The tail can be added so it rings out.
Compressor	Shift+C	Turns down what is above the threshold.
Limiter	Shift+T	Stops peaks from going above the ceiling. Looks ahead, so nothing gets through.

Control	Shortcut	What it does
Noise gate	Shift+U	Silences what is below the threshold.

Tool bar



Control	Shortcut	What it does
New empty sound	Alt+N	Starts with an empty sound. Undo brings the old one back.
Open file	Ctrl+O	Opens an audio file. You can also drop a file on the display.
Save as WAV	Ctrl+S	Saves the sound as a WAV file. In this web mockup the browser may block the download.
Undo	Ctrl+Z	Nothing to undo.
Redo	Ctrl+Y	Nothing to redo. Ctrl+Shift+Z repeats the last operation.
History	Alt+Y	Every step of this sound, with its own undo that is separate from the host. Click to jump back to any step.
Cut	Ctrl+X	Moves the selection to the clipboard and closes the gap.
Copy	Ctrl+C	Copies the selection to the clipboard.
Paste	Ctrl+V	Puts the clipboard at the cursor, or over the start of the selection.
Delete	Del	Deletes the selection and closes the gap.
Crop	T	Keeps only the selection and throws the rest away.
Select all	Ctrl+A	Selects the whole sound.
Snap to zero	Z	Moves the selection edges to the nearest zero crossing, so cuts do not click.
Zoom in	+	Shows less of the sound, in more detail.
Zoom out	-	Shows more of the sound.
Show everything	F	Fits the whole sound in the display.
Zoom to selection	S	Fills the display with the selection.
Repeat last operation	R	Nothing to repeat yet. Apply an operation first.
Grab the last seconds	G	Keeps what you already heard, even though you were not recording. Uses the pre-roll length.
Recording in stereo	Alt+O	What a new take is made of. Click to switch. The display shows one lane for a mono sound and two for a stereo one. Operations like Make mono change a sound you already have.
Input: mixer channel	M	Where the recording comes from. In the plugin: the mixer channel it sits on. Here: a demo loop stands in, or your microphone. Click to switch.
Monitoring the input	Alt+M	Lets you hear what comes in while you record. Mute it when the sound already plays somewhere else.
Spectral view		Shows the sound as a picture of its frequencies, with tools to cut, erase, copy and amplify parts of it.

Readout and sound info



Control	Shortcut	What it does
Channels		What the sound is made of, as detected. Click to make it mono (mix, left only or right only), swap the sides or split it into mid and side.
Sample rate		The sample rate of this sound, as detected. Click to convert it to another rate, or to change the rate without converting.
Bit depth		The bit depth of this sound and of the file you save. Click to change it. Going lower changes the sound, with dither if you want it.

Control	Shortcut	What it does
Time line unit		How positions and the time line are shown: time, seconds, samples, or bars and beats.
Sample tempo		The tempo Meowcorder found in this sound. Click to set it yourself, double it, halve it, or say how many bars the sound is.
Sync		Free plays the sound at its own speed. Sync plays it at the host tempo. Click to choose, and to pick the warp mode.
Warp settings		Opens the warp panel: tempo sync, warp mode, transpose, fine tune and formants.

Level meters



Control	Shortcut	What it does
Level meters and trigger arrow		Input (left bar) and output (right bar) level in dBFS, with peak hold. The yellow arrow is the record trigger level, the same value as the Threshold knob: drag it up or down. It starts a recording in On input mode.

Display, views and time line



Control	Shortcut	What it does
Live input		Input, always listening: the last 3 seconds. The pink band is the pre-roll that a recording would include.
Waveform	Alt+W	Shows the sound as a wave. Drag to select a time range. The usual way to cut, copy and process. Click to show it alone.
Spectral view	Alt+G	Shows the sound as a picture of its frequencies, with tools to cut, erase, copy and amplify parts of it. Click to show it alone.
Pitch view	Alt+K	A piano roll of the notes found in the sound. Drag a note up or down to change its pitch, and the sound follows. Click to show it alone.

Control	Shortcut	What it does
Show views together	Alt+B	Turn on to see more than one view at once. Overlay puts them on top of each other, Split stacks them. The view In front is the one your tools work on.
Waveform pencil	Shift+Q	Draw the wave by hand. Zoom in until you can see single samples, then drag over the wave to set them. Shift+Q turns it on or off.
Mouse	Shift+1	Click for the cursor, drag to select a time range.
Selector	Shift+2	Drag a box over time and frequency. Use Cutter, Amp or Copy on the box from the Spectral menu.
Scissors	Shift+3	Drag across time: that slice is cut out and the gap closes.
Cutter	Shift+4	Drag a box: the sound inside it is removed, the length stays.
Eraser	Shift+5	Paint over what should go. It is removed when you let go.
Copy	Shift+6	Drag a box to copy that piece, then click with Paste where it should land.
Paste	Shift+7	Click where the copied piece should go. Left edge is time, height is frequency.
Amp	Shift+8	Drag a box: it gets the level change set next to the tools.
Pencil	Shift+9	Draw a line on the picture: a pure tone follows your pen, so you can draw new sounds. Drag forward in time. The Pen level is set next to the tools. Click for a short beep.
Amp level		Level change the Amp tool applies to the box.
Pen level		Level of the tone the Pencil draws.
Eraser brush		Size of the Eraser brush.
Snap notes to semitones		Moves every note to the nearest exact semitone. Like a tuner for the whole sound, or for the selection.
Formants follow		The voice character moves with the pitch, like speeding a tape.
Keep formants		Keeps the voice character (the formants) while the pitch changes. Best for voices and instruments.
Drag mode		How a dragged note lands.
Waveform display		The sound, stereo or mono. Drag to select, click to place the cursor, Shift click to extend, wheel to zoom, Alt wheel to scroll. Operations work on the selection, or on everything when nothing is selected.
Spectral display		Spectral view. Horizontal: time. Vertical: frequency, 20 Hz at the bottom, log scale. Colour is level. Pick a tool above it to change the sound by looking at it.
Pitch display		Pitch view, like a piano roll: the notes found in the sound. Drag a note up or down to change its pitch.
Time line		Time line. Click to place the cursor. The unit (time, seconds, samples, bars and beats) is set in the bar above the display.
Whole sound		The whole sound. The lighter box is what you see above. Drag it to scroll, click to jump, wheel to zoom.

Recording

- Now starts at once. On input waits for sound: recording begins when the level passes the Threshold. Count-in plays a click for 1, 2 or 4 bars at the Tempo, then records.
- The arrow in the meter is the trigger level, the same value as the Threshold knob. Drag it up if background noise starts the recording too early, and down if quiet starts are missed. It only does something in On input mode.
- Pre-roll adds the sound from just before the trigger, so the start of a sound is not cut off. The pink band on the live strip shows how much.
- A new take can Replace the sound, Append at the end, Punch over the selection, or Insert at the selection.
- Grab the last seconds keeps what you already heard even though you were not recording.

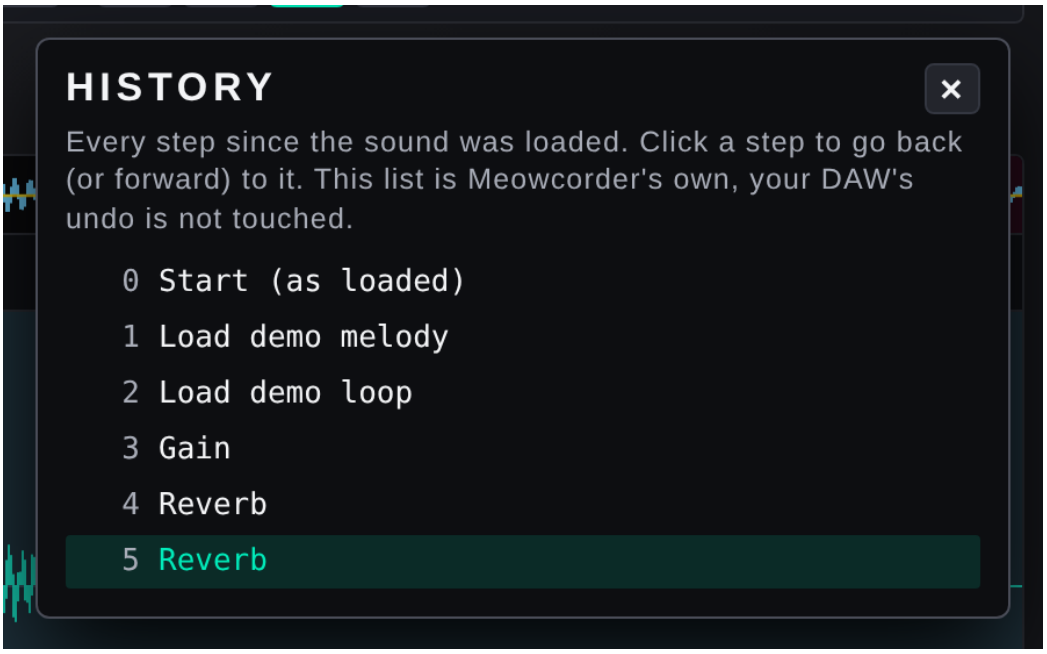
- Record in mono makes new takes mono. The input can be the mixer channel the plugin sits on, or a microphone in the standalone app.



Waveform view with a selection. The arrow in the meter on the right is the record trigger level.

Editing and history

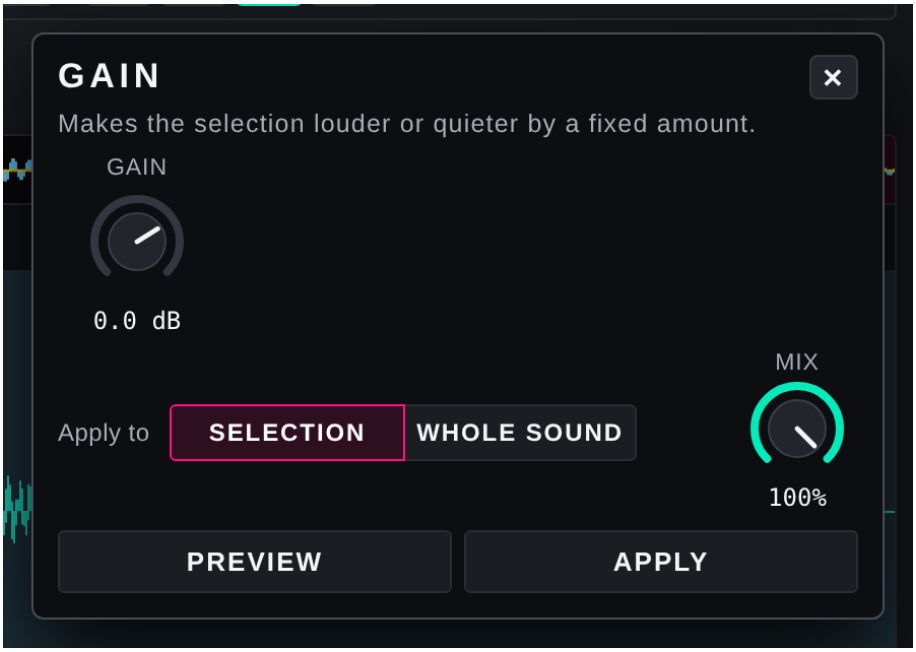
- Click places the cursor, drag selects, Shift click extends the selection, double click selects everything. The wheel zooms and Alt+wheel scrolls. The strip at the bottom shows the whole sound, drag its light box to scroll.
- Cut, copy, paste, delete, crop and Snap to zero crossing work on the selection.
- Meowcorder keeps its own list of steps. Open History (clock button) to see every step since the sound was loaded and click any step to go back to it, or forward again.
- Ctrl+Shift+Z redoes a step. When there is nothing to redo it applies the last action again, so you can press it over and over to repeat the same operation.



History: click any step to go back to it, or forward again. This list is Meowcorder's own; your DAW's undo is not touched.

Operations

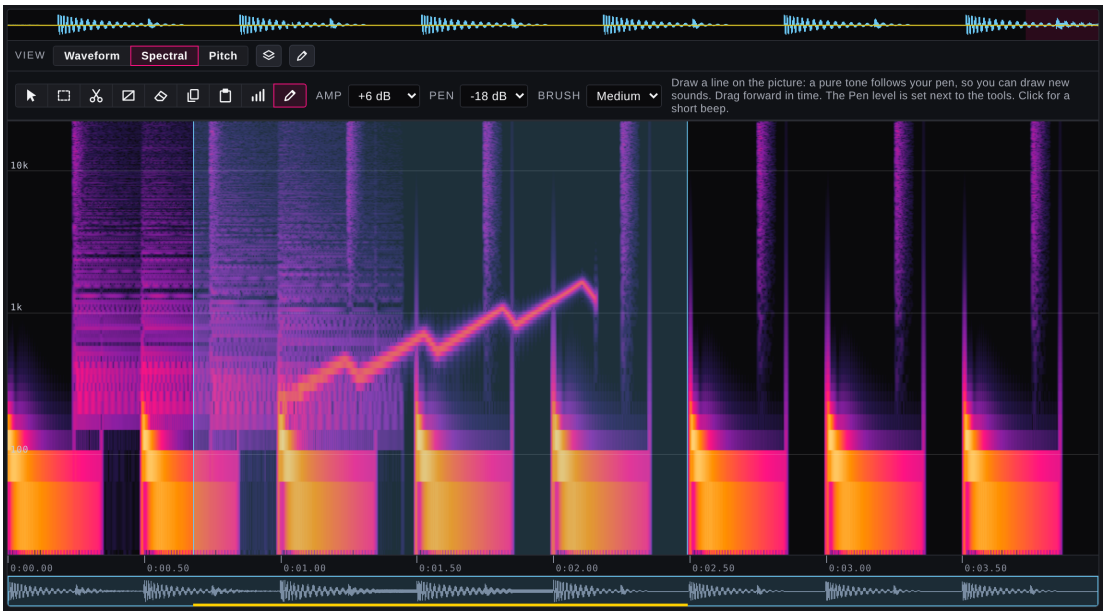
- 24 operations: gain, normalize, fade in and out, reverse, invert, DC remove, silence, make mono, balance, pitch and speed, bitcrusher, Drive, EQ, filter, delay, chorus, phaser, reverb, compressor, limiter, noise gate, trim silence, make loop.
- Each one opens a popup with its knobs, a Mix knob for how much of the effect is added, and an Apply to switch: Selection (the default) or Whole sound.
- Preview plays the result without changing the sound. Apply makes the change and adds a step to the history.



An operation popup: knobs, Mix, and Apply to Selection or Whole sound.

Views

- Three views: Waveform, Spectral and Pitch. By default you see one at a time, and the one you see is the one your tools work on.
- The layers button shows several together. Overlay puts them on top of each other, Split stacks them. In front says which view gets your edits. The others follow live: remove the low end in the spectral view and the waveform below it gets thinner at that spot.
- Spectral view: frequency goes up, 20 Hz at the bottom and a log scale, time goes right, colour is level. Tools: Mouse, Selector, Scissors, Cutter, Eraser, Copy, Paste, Amp and Pencil. The Eraser works while you paint.
- Pitch view: a piano roll of the notes Meowcorder found. Drag a note up or down to change its pitch, double click a note to snap it to the nearest semitone, or use Snap notes to semitones for all of them. Choose whether the formants follow the pitch or are kept.



Spectral view with the Pencil: the drawn line is real sound added to the file.



Pitch view: the notes found in a melody, ready to drag.



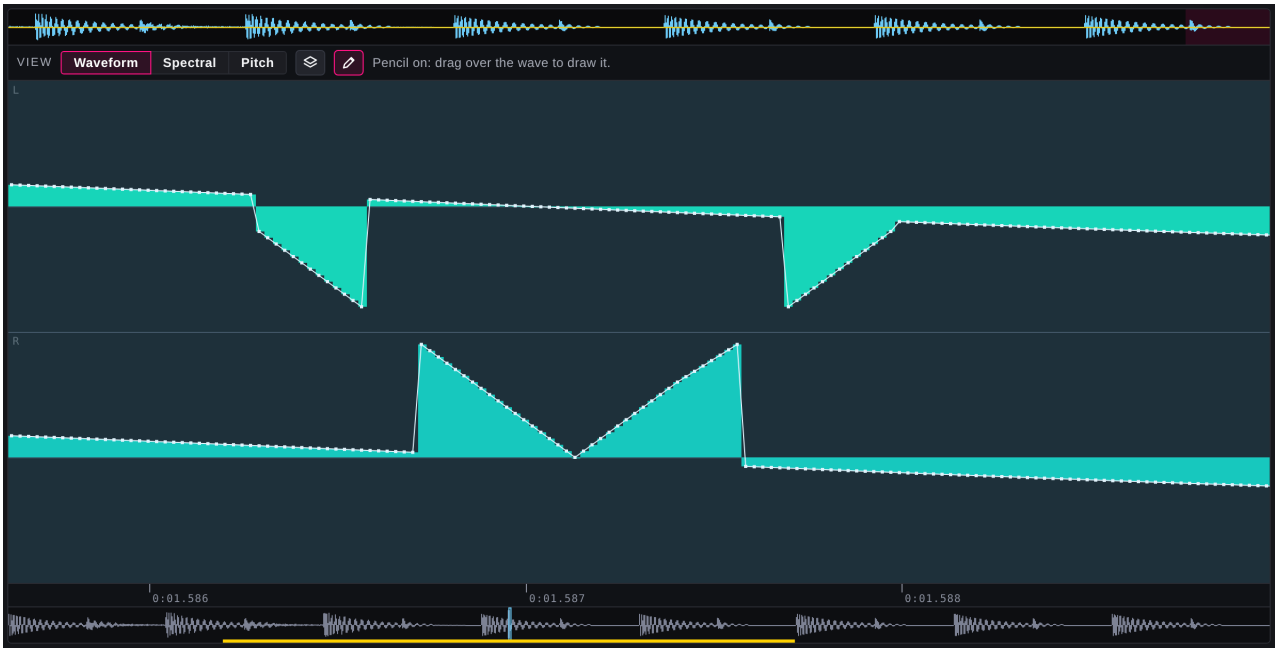
Split: Waveform, Spectral and Pitch stacked. The view In front gets your edits.



Overlay: the views on top of each other.

Pencils

- Spectral pencil: pick Pencil in the spectral tools and drag forward in time. A pure tone follows your pen and is added to the sound at the Pen level (-36 to -6 dB). A single click adds a short beep. The waveform updates while you draw.
- Waveform pencil: press Shift+Q or the pencil button in the view bar. It zooms in to single samples. Drag over the wave to draw it. In a stereo sound the lane under the pen is the side that changes. It only draws when you can see single samples; zoom in if it says so.
- Each stroke is one step in the history, so Ctrl+Z takes it back.



Waveform pencil at sample level: the drawn wave replaces the samples under the pen.

Timing and warp

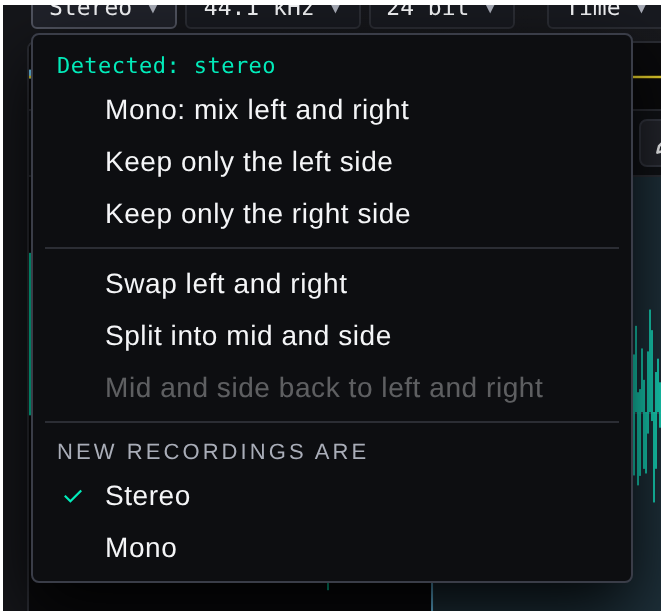
- The time line can show time, seconds, samples, or bars and beats (the Time button in the info bar).
- Meowcorder listens for the tempo of the sound (the bpm button). Click it to set the tempo yourself, double it, halve it, or say the sound is exactly 1, 2, 4 or 8 bars.
- Free plays the sound at its own speed. Sync to host plays it at the project tempo. Warp modes: Stretch, Repitch, Beats and Texture.
- Transpose, Fine tune and Formant change the pitch without changing the length. Bake in puts the warp into the sound; until then it only affects playback.



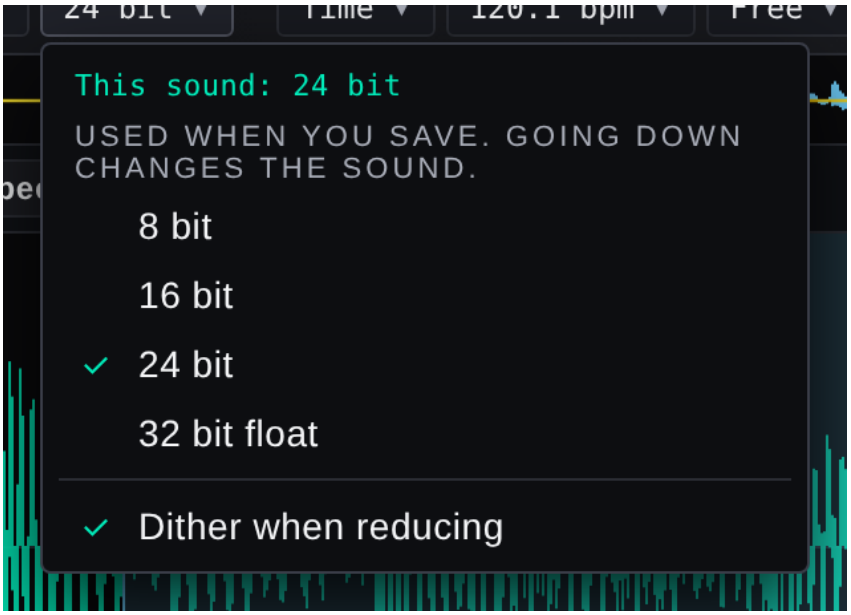
The warp panel: sync, warp mode, transpose, fine tune, formants, and Bake in.

Sound info

- The Channels, kHz and bits buttons show what Meowcorder found in the sound. Click them to change it.
- Channels: make it mono by mixing both sides, keep only the left or only the right, swap the sides, or split it into mid and side and back.
- kHz: convert to another sample rate (same sound, new rate), or set the rate only, which changes speed and pitch like a tape.
- Bits: the bit depth used when you save. Going lower changes the sound, with dither if you want it.

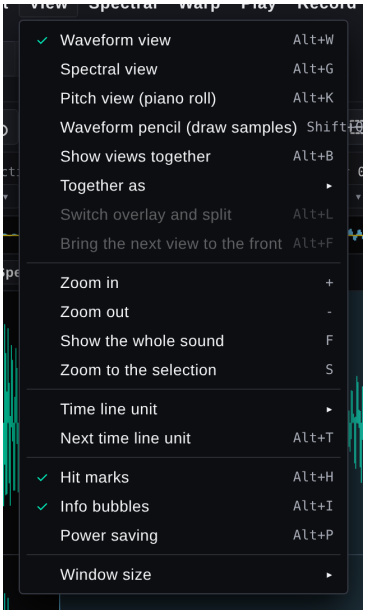


The Channels button shows what was detected and offers the changes.

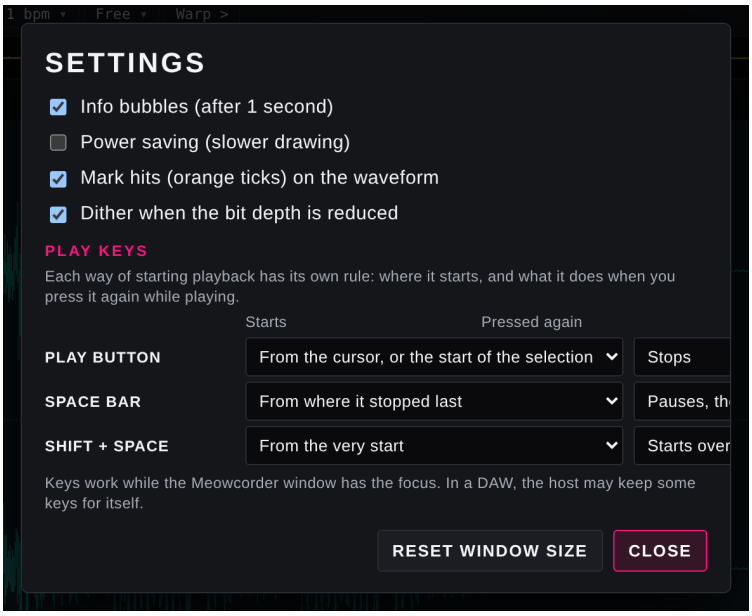


The bit depth button, with dither for going lower.

Menus and settings



The View menu. Every menu shows its shortcuts.



Settings, including the Play keys.

Play keys

- Space and Shift+Space play and stop. What each does is yours to choose in Settings under Play keys, for example start from the beginning, resume where it stopped, or play from the cursor.
- Shortcuts work while Meowcorder has focus. Your DAW may keep some keys for itself; every action is also in the menus.

Defaults: Space = resume then pause; Shift+Space = start then restart; the Play button = cursor then stop.

Keyboard shortcuts

File

Action	Shortcut
New empty sound	Alt+N
Open file...	Ctrl+O
Save as WAV	Ctrl+S
Audio on or off	Alt+A

Edit

Action	Shortcut
Undo	Ctrl+Z
Redo	Ctrl+Y
Redo, or repeat the last operation	Ctrl+Shift+Z
Cut	Ctrl+X
Copy	Ctrl+C
Paste	Ctrl+V
Delete selection	Del
Crop to selection	T
Repeat last operation	R
History...	Alt+Y

Select

Action	Shortcut
All	Ctrl+A
None	Esc
From cursor to the end	Shift+End
From the start to the cursor	Shift+Home
Snap edges to zero crossings	Z

View

Action	Shortcut
Waveform view	Alt+W
Spectral view	Alt+G
Pitch view (piano roll)	Alt+K
Waveform pencil (draw samples)	Shift+Q
Show views together	Alt+B
Switch overlay and split	Alt+L
Bring the next view to the front	Alt+F
Zoom in	+
Zoom out	-
Show the whole sound	F
Zoom to the selection	S
Next time line unit	Alt+T
Hit marks	Alt+H
Info bubbles	Alt+I
Power saving	Alt+P
75%	Alt+1
100%	Alt+2
125%	Alt+3
150%	Alt+4
200%	Alt+5
Back to 100%	Alt+0

Spectral

Action	Shortcut
Mouse	Shift+1
Selector	Shift+2
Scissors	Shift+3
Cutter	Shift+4
Eraser	Shift+5
Copy	Shift+6
Paste	Shift+7
Amp	Shift+8
Pencil	Shift+9
Cut the box	Alt+X
Amp the box	Alt+U
Copy the box	Alt+C
Paste at the cursor	Alt+V

Warp

Action	Shortcut
Warp, tempo and pitch...	Alt+Q
Sync to the host tempo	Alt+S
Bake the warp into the sound	Alt+E
Snap the notes to semitones	Alt+D

Play

Action	Shortcut
Play, Space bar rule	Space
Play, Shift+Space rule	Shift+Space
Loop	L
Preview the chosen operation	P
Cursor to the start	Home
Cursor to the end	End

Record

Action	Shortcut
Record or stop	Enter
Grab the last seconds	G
Monitor the input	Alt+M
Record in mono	Alt+O
Now	1
On input (when sound arrives)	2
After a count-in	3
Replace the sound	4
Append at the end	5
Punch over the selection	6
Insert at the selection	7
Use the microphone instead of the mixer channel	M

Process

Action	Shortcut
Apply the chosen operation	Ctrl+Enter
Gain	Shift+G
Normalize	Shift+N
Fade in	Shift+I
Fade out	Shift+O
Reverse	Shift+R
Invert polarity	Shift+P
Remove DC offset	Shift+D
Silence	Shift+S
Trim silence	Shift+X
Make mono	Shift+M
Balance	Shift+B
Pitch and speed	Shift+Y
Make loop	Shift+J
Bitcrusher	Shift+K
Drive	Shift+V
EQ	Shift+E
Filter	Shift+F
Delay	Shift+L
Chorus	Shift+H
Phaser	Shift+A
Reverb	Shift+W
Compressor	Shift+C
Limiter	Shift+T
Noise gate	Shift+U

Help

Action	Shortcut
Guide	F1
Keyboard shortcuts	?
Settings	Ctrl+,

Install

Installers are not built yet; they come with the first plugin release (Windows installer, macOS package, Linux Flatpak). Until then the interactive mockup runs in any browser: open docs/mockups/Meowcorder.html. Everything is made inside the page and nothing is sent anywhere.

Under the hood

- The editing engine is plain C++17, tested by 619 checks. The mockup uses a JavaScript copy of it, and a parity test checks that all 24 operations give the same result in both.
- Spectral edits use a Hann window of 2048 samples with a hop of 512. The spectral pencil adds a sine wave with 3 ms fades at each end.
- Warp, pitch shifting, tempo detection, note detection, rate conversion and bit depth reduction are JavaScript in the mockup for now (28 checks plus the pencil checks) and move to C++ with the plugin.
- Warp preview renders the sound first, so a long sound may pause for a moment.
- Files: WAV in at 8, 16, 24 and 32 bit (integer and float), WAV out at 8, 16, 24 and 32 bit float.
- Meowcorder collects nothing and makes no network calls.

Support

Free and open source. If you like it, you can support it on Ko-fi at ko-fi.com/studio87. Nothing is locked and nothing is collected.